

Physical Chemistry (II)

Lecture 03

Principles of Quantum Mechanics



In the Last Class

- Classical mechanics
 - Physics of particles
 - Physics of waves
- Old quantum theory
 - Wave-particle duality
 - Bohr atom: quantization
 - Introduction to spectroscopy

Physics of Particles

- Described by $\vec{r}$ and $\vec{v}$ (or equivalently $\vec{p} = m\vec{v}$)
- Kinetic energy $T = mv^2/2$ + potential energy V
- Dynamics: $\vec{F} = m\vec{a}$ or

$$\frac{\partial^2 \vec{r}}{\partial t^2} = -\frac{\nabla V}{m}$$

Physics of Waves

- Described by amplitude $A = A(\vec{r}, t)$
- Dynamics: wave equation

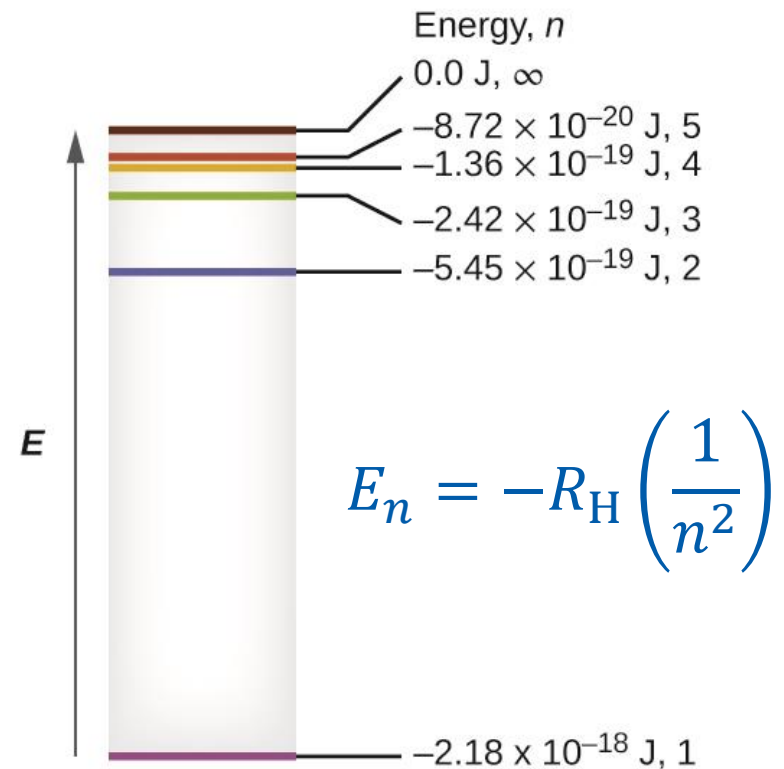
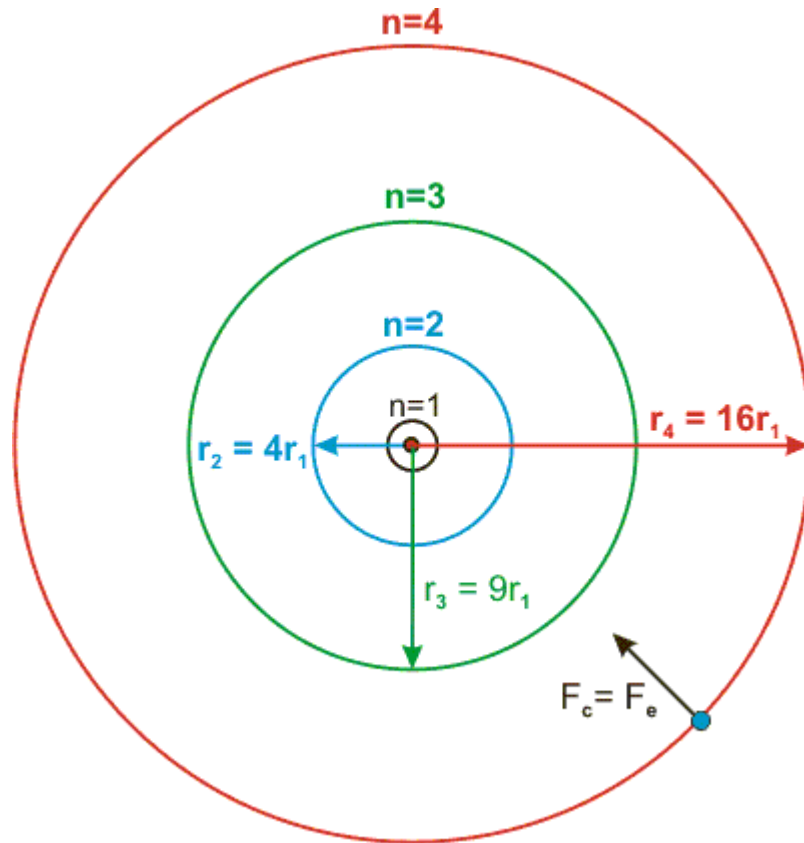
$$\frac{\partial^2 A}{\partial t^2} = u^2 \nabla^2 A$$

- **Superposition of waves:** $A(\vec{r}, t) = A_1(\vec{r}, t) + A_2(\vec{r}, t) + \dots$
- Refraction, interference, diffraction

Wave-Particle Duality

- Particle nature of light
 - Photoelectric effect: $E = h\nu = hc/\lambda$
 - Compton scattering: $p = h/\lambda$
- Wave nature of matter
 - Matter wave: $\lambda = h/p = h/(mv)$
- Wave-particle duality
 - Light and matter exhibit both properties of waves and particles
 - Bohr's complementarity principle

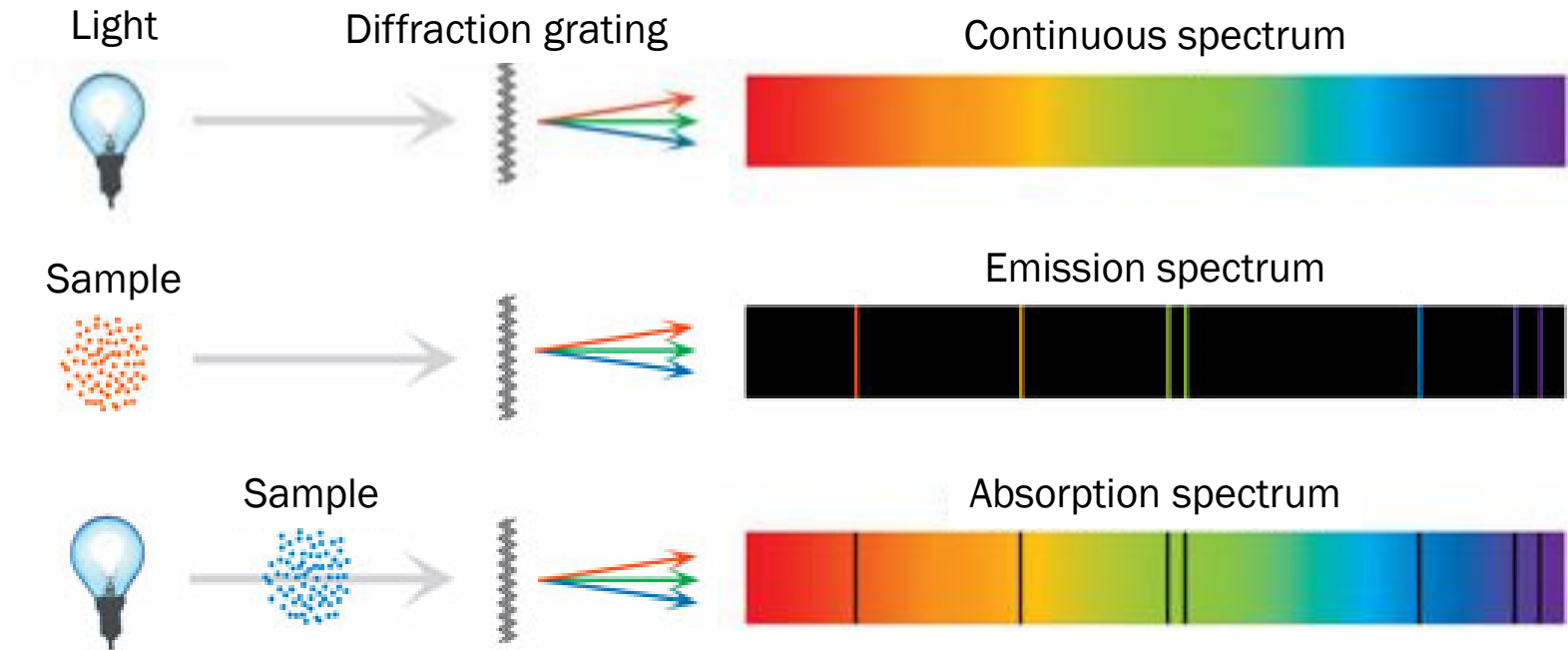
Bohr Atomic Model



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Spectroscopy

Study of the absorption and emission of **light** by **matter**



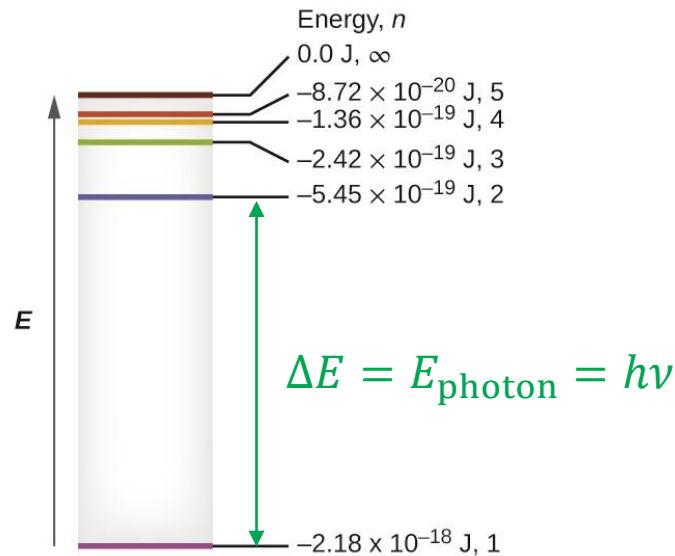
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Why Do We Get “Lines”?

- Quantization of light

$$E = h\nu$$

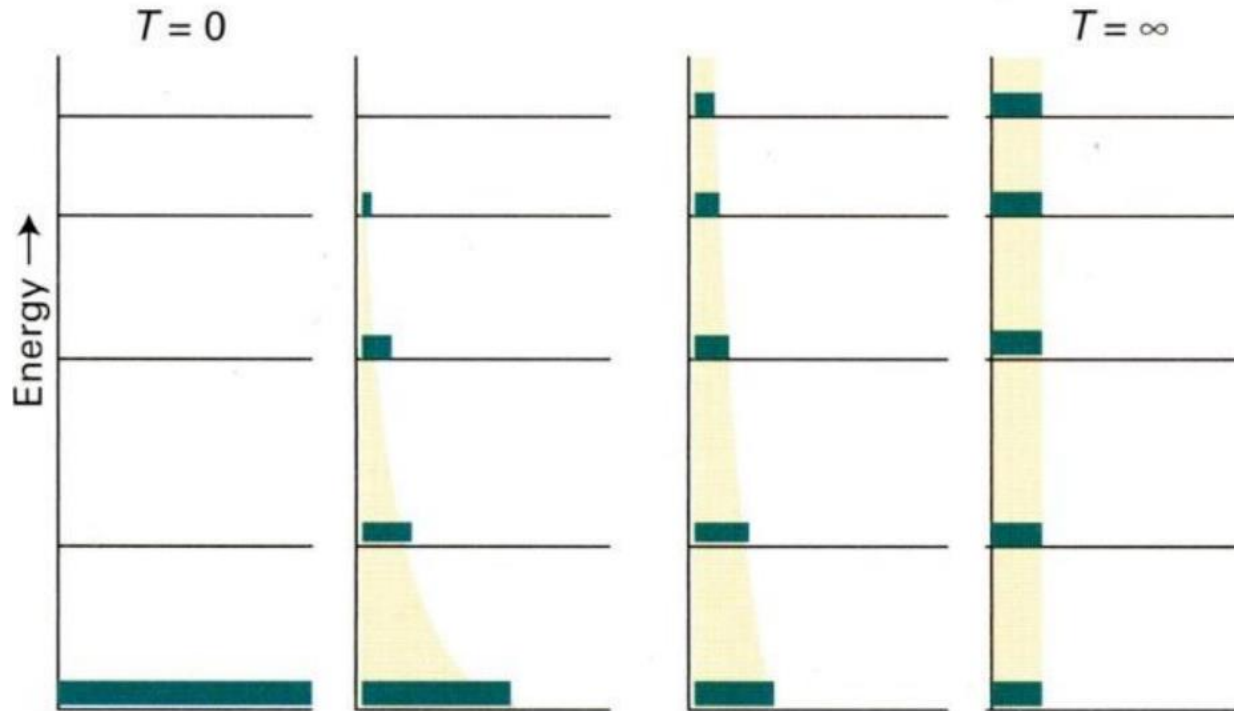
- Quantization of atomic energy



(Image: <https://opentextbc.ca/chemistry/chapter/6-2-the-bohr-model/>)

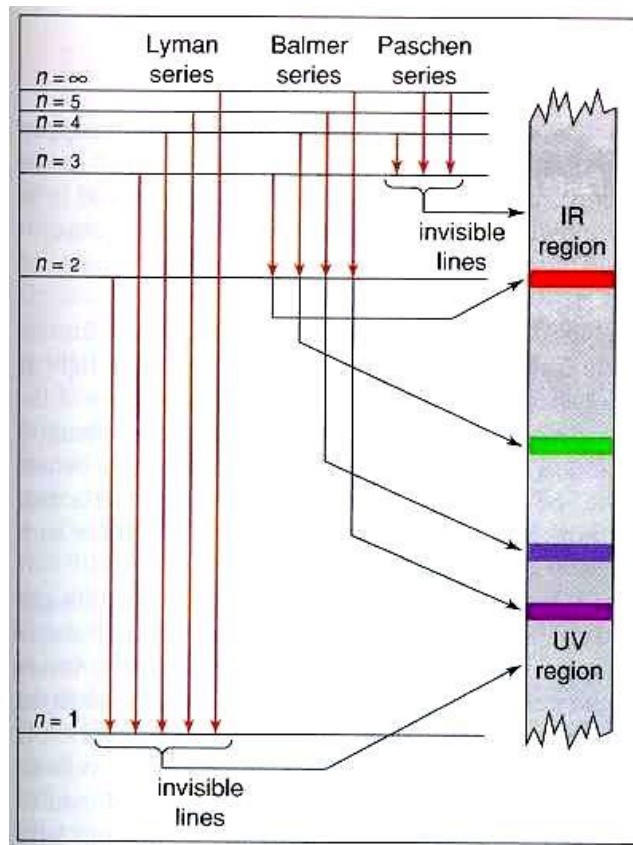
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Thermal Excitation



$$P(E) \propto e^{-E/k_B T} \text{ (Boltzmann factor)}$$

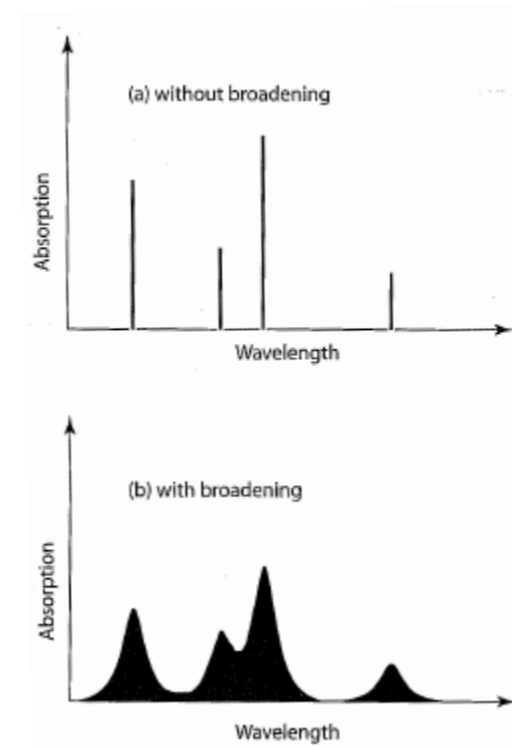
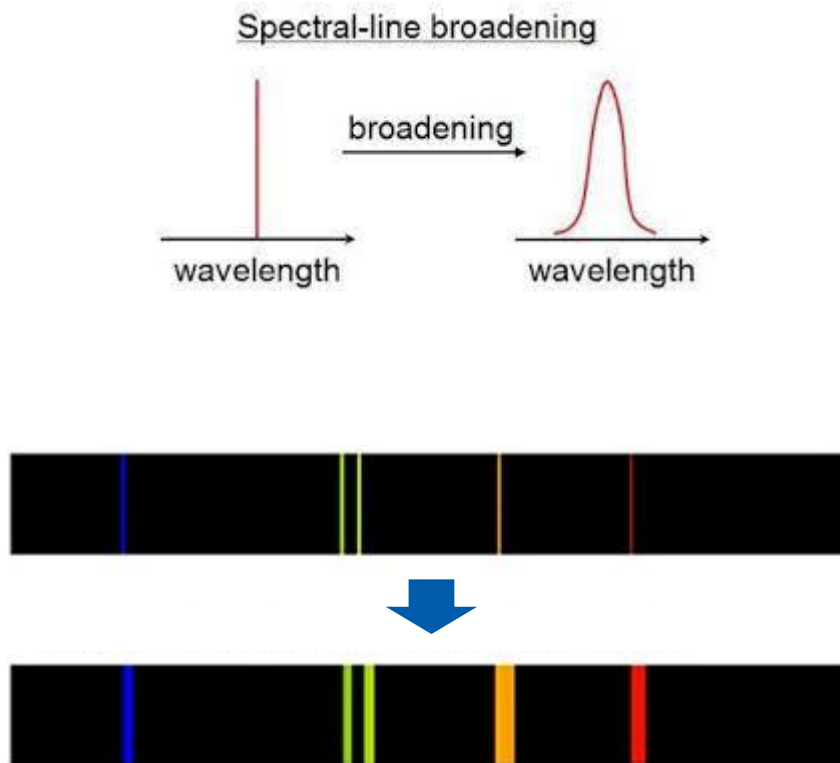
Transitions in H Atom



(Image: <https://bramblechemistry.weebly.com/4e2-bohr-model.html>)

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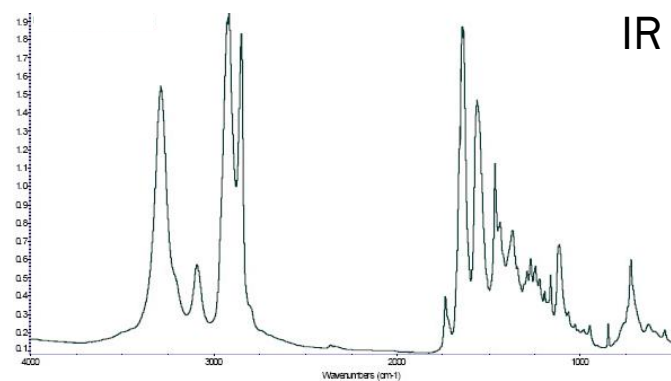
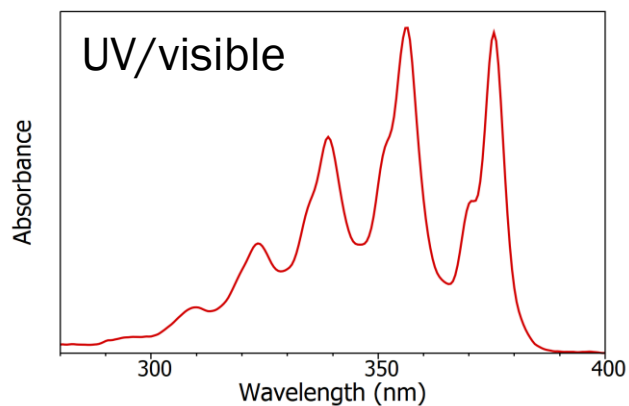
They Are NOT Lines Though...



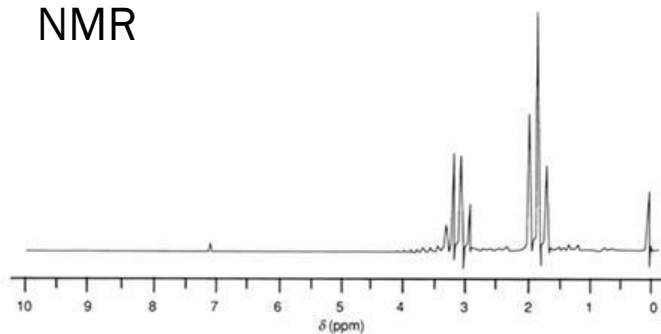
(Image: https://uomustansiriyah.edu.iq/media/lectures/6/6_2018_12_22!12_29_47_AM.pdf
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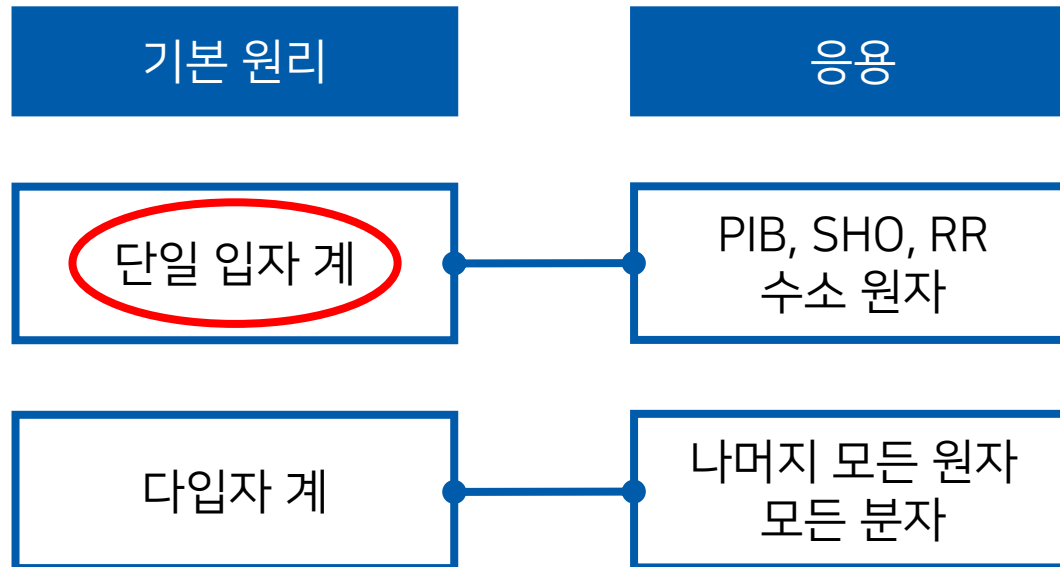
Real Spectra



NMR



Where Are We?

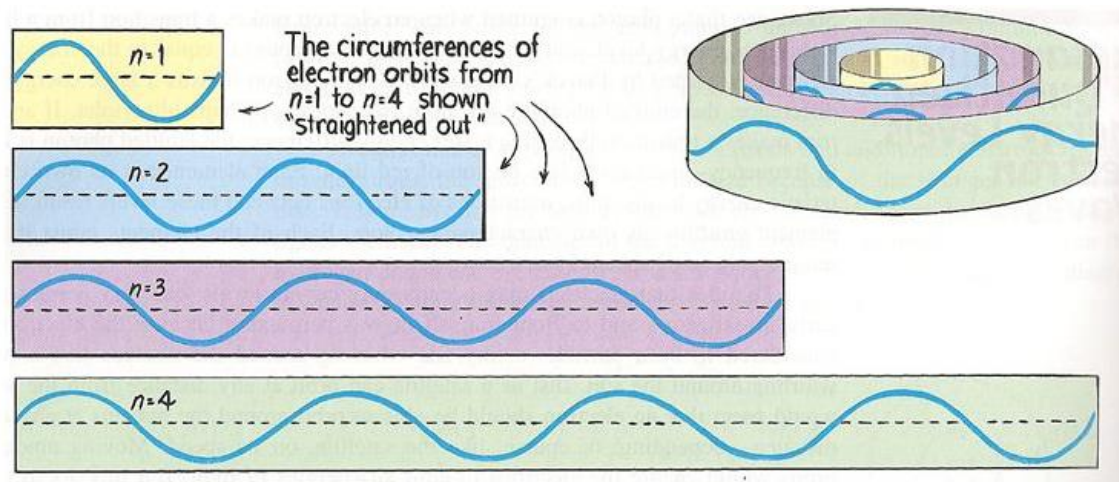


Postulates of QM

1. Wavefunction and its interpretation
2. Quantum measurement
 - Observables and operators
 - Measurement and collapse of wave function
 - Uncertainty principle
3. Time evolution of wavefunction: Schrödinger equation

Wavefunction

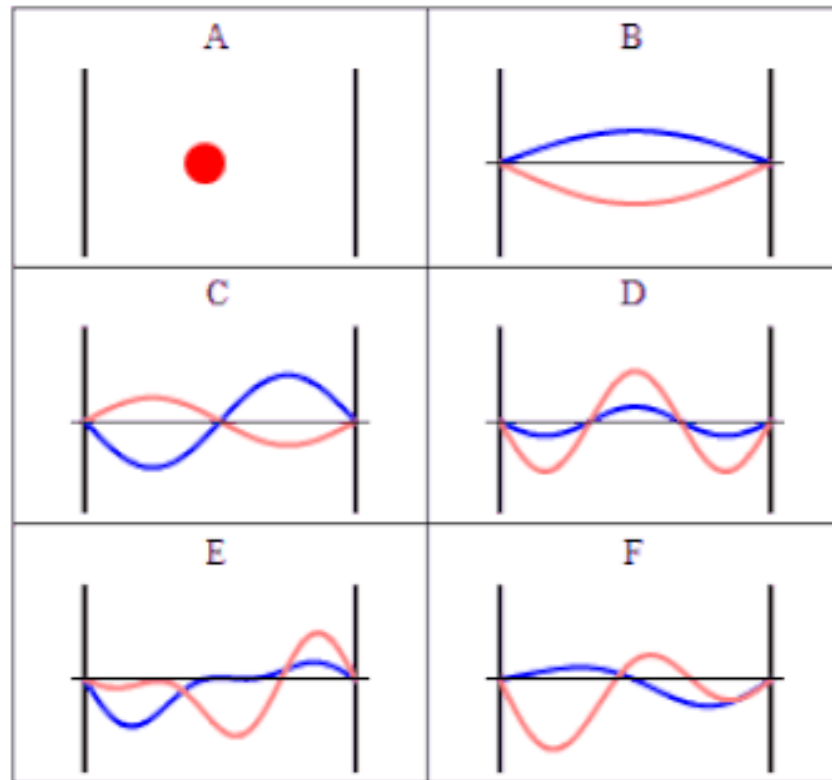
The state of a quantum-mechanical particle is represented by a **certain complex function** that behaves like a **wave**



(Image: <https://chemistry.stackexchange.com/questions/116840/why-do-solutions-of-electron-in-a-box-and-in-a-ring-predict-coefficients-for-l>)

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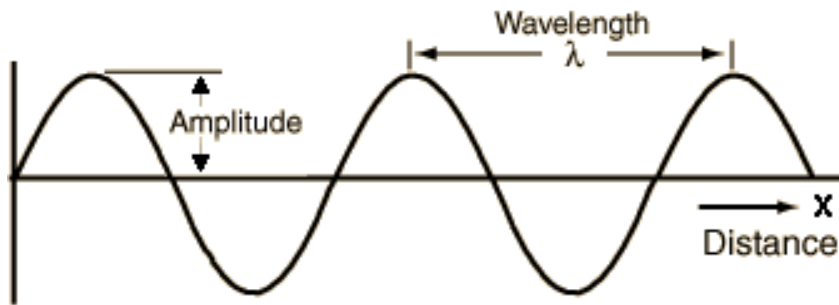
Example of a Wavefunction



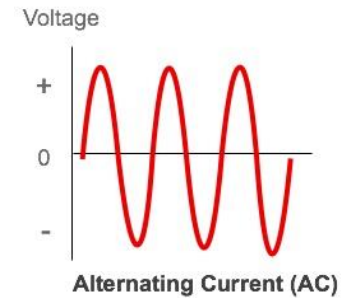
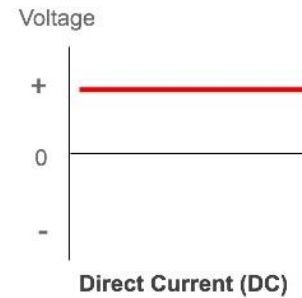
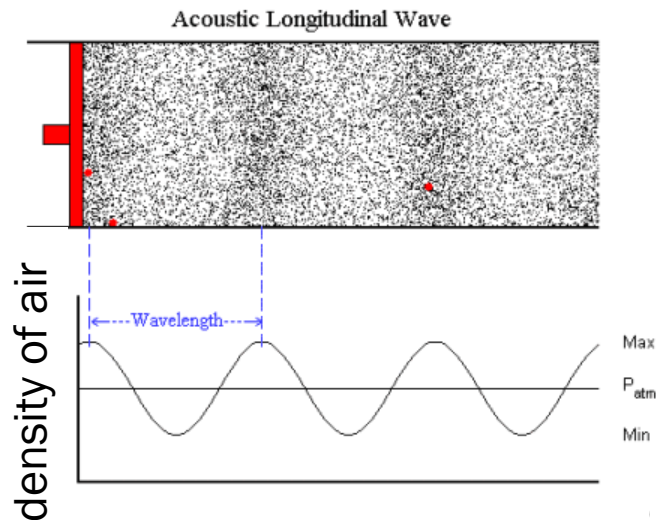
(Image: <https://commons.wikimedia.org/wiki/File:InfiniteSquareWellAnimation.gif>)

“Amplitude” of a Wave

Some value assigned for each point of the space



0 +1 +2 +1 0 -1 -2 -1 0 +1 +2 +1



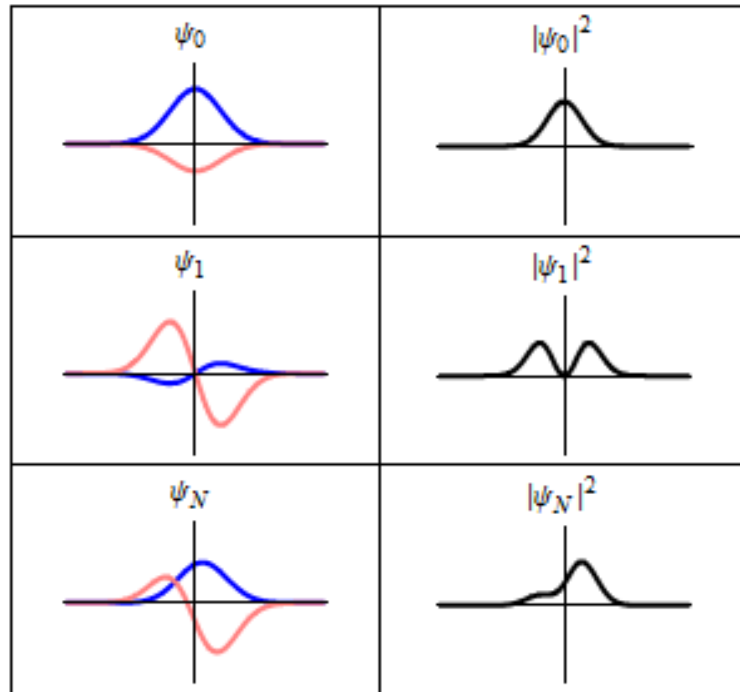
“Amplitude” of a Wavefunction

For a wavefunction $\Psi(x, t)$,
its square $|\Psi(x, t)|^2$ gives the **probability**
of finding the particle at point x at time t
(Born’s statistical interpretation)

What really matters is the **PROBABILITY!**

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Wavefunction and Probability



Normalization

The integral of the probability must be one:

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 1$$

(Modified) Example 4-1

A wave function has the form of

$$\Psi_n(x, t) = \left(\frac{2}{a}\right)^{1/2} \sin\left(\frac{\pi x}{a}\right) e^{-2\pi i E t / \hbar}, \quad 0 \leq x \leq a.$$

Show that the wave function is normalized.

The probability density is

$$|\Psi_n(x, t)|^2 = \Psi_n(x, t) \Psi_n^*(x, t) = \frac{2}{a} \sin^2\left(\frac{\pi x}{a}\right)$$

The normalization condition leads to

$$\int_{-\infty}^{\infty} |\Psi_n(x, t)|^2 dx = \frac{2}{a} \int_0^a \sin^2\left(\frac{\pi x}{a}\right) dx = \frac{2}{a} \times \frac{a}{2} = 1$$

RS 2.2
M 4.1

Once We Know a Wavefunction...

A wavefunction contains all information
about the state of the system

Quantum-mechanical measurement?

Observables and Operators

- Every **CM observable** has a corresponding **QM operator**
- An operator performs a mathematical operation on the following function
 - e.g. “+3” operator, “ $\times 2$ ” operator, “ d/dx ” operator
 - Typically denoted by a letter with a **caret** ($\hat{}$)

Observables and Operators

- Every **CM observable** has a corresponding **QM operator**

- Position operators: $\hat{x} = x$, $\hat{y} = y$, $\hat{z} = z$

- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \quad \hat{p}_y = \frac{\hbar}{i} \frac{\partial}{\partial y}, \quad \hat{p}_z = \frac{\hbar}{i} \frac{\partial}{\partial z}$$

$\hbar/2\pi$
↓
○

- System energy operator (Hamiltonian):

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(\hat{x}, \hat{y}, \hat{z})$$

Eigenvalue Equation

For a given operator $\hat{A}$, a function f such that

$$\hat{A}f = af$$

is called an **eigenfunction** of $\hat{A}$, and a is called an **eigenvalue**. This equation is called an **eigenvalue equation**.

e.g. For $\hat{A} = d/dx$,

e^{2x} is an eigenfunction with the eigenvalue of 2.

Properties of QM Operators

- Eigenvalues are **real**
- Eigenfunctions form a **complete** set

“Any arbitrary function can be represented as a linear combination of the functions from a complete set”
- Eigenfunctions are **orthogonal** to each other

“Two functions are orthogonal to each other if their inner product is zero”

$$\int f^* g \, d\tau = 0$$

Measurement in QM

If a wavefunction Ψ of the system satisfies

$$\hat{A}\Psi = a\Psi$$

for a certain operator $\hat{A}$, the **measurement** of the corresponding observable always gives a (eigenvalue).

Example

The x -axis momentum operator is

$$\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}.$$

If a wavefunction of the system is

$$\Psi(x) = e^{ikx},$$

the following eigenvalue equation holds:

$$\hat{p}_x \Psi(x) = (\hbar k) \Psi(x).$$

Hence, measurement of the system momentum always gives $\hbar k$.

M 4.8

If Wavefunction $\neq$ Eigenfunction?

If a wavefunction is represented as a **linear combination**

of the eigenfunctions of an operator ($\hat{A}f_k = a_k f_k$),

$$\Psi = c_1 f_1 + c_2 f_2 + c_3 f_3 + \cdots + c_n f_n,$$

the measurement gives the eigenvalue a_i with

the **probability** of the squared coefficient $|c_i|^2$.

Example

The x -axis momentum operator $\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}$:

eigenfunction e^{ikx} with eigenvalue $\hbar k$

If a wavefunction of the system is

$$\psi(x) = \frac{\sqrt{2}}{\sqrt{3}} e^{ix} + \frac{1}{\sqrt{3}} e^{-2ix},$$

the measurement of the system momentum gives

- $\hbar$ with the probability of $2/3$; and
- $-2\hbar$ with the probability of $1/3$

Collapse of a Wavefunction

When you obtain a certain value from measurement, the wavefunction becomes the **corresponding eigenfunction**.

e.g. For $\Psi(x) = \frac{\sqrt{2}}{\sqrt{3}} e^{ix} + \frac{1}{\sqrt{3}} e^{-2ix}$,

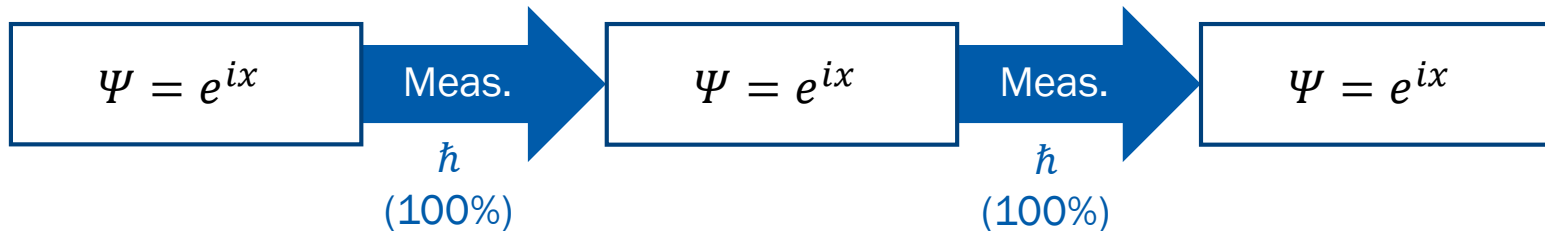
if the momentum measurement gives

- $\hbar \rightarrow$ the wavefunction becomes $\Psi(x) = e^{ix}$
- $-2\hbar \rightarrow$ the wavefunction becomes $\Psi(x) = e^{-2ix}$

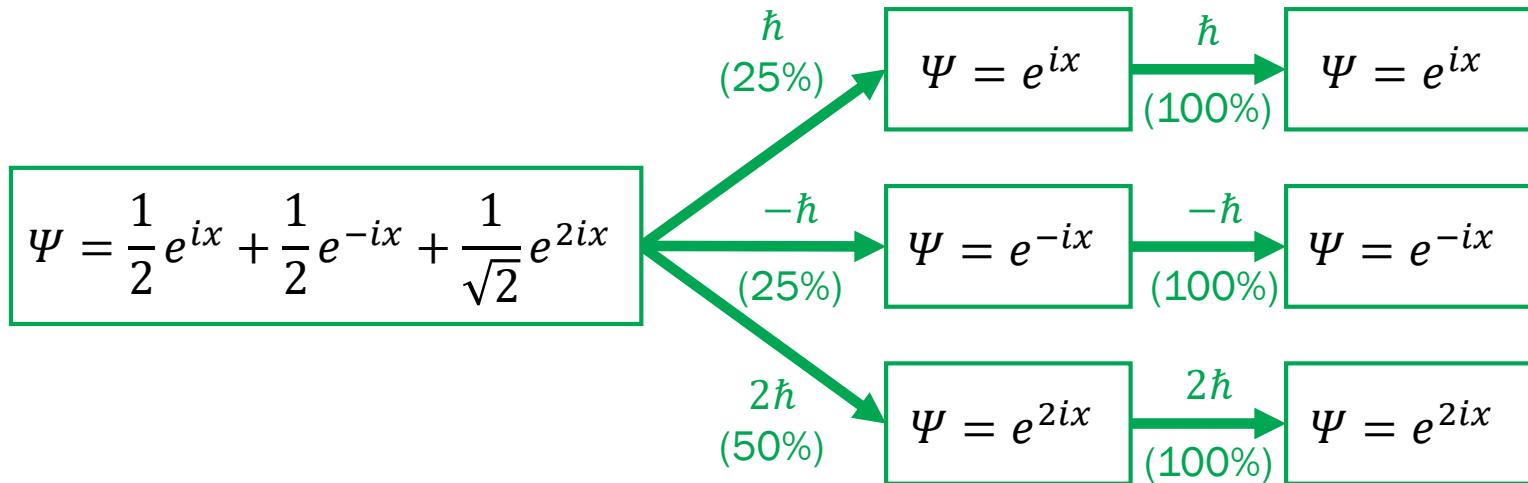
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Quantum Measurement

- If the wavefunction is the eigenfunction of the operator

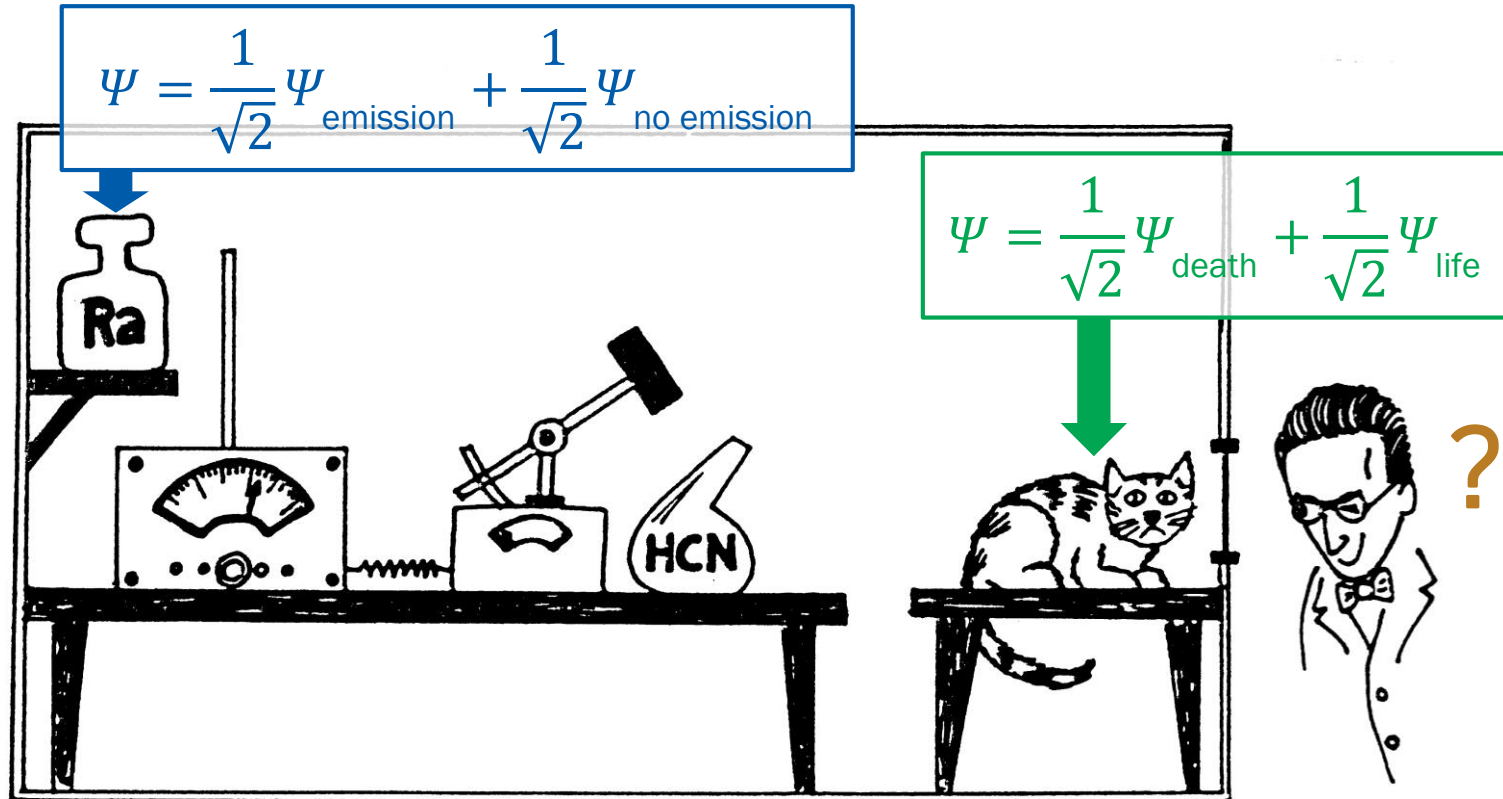


- Otherwise



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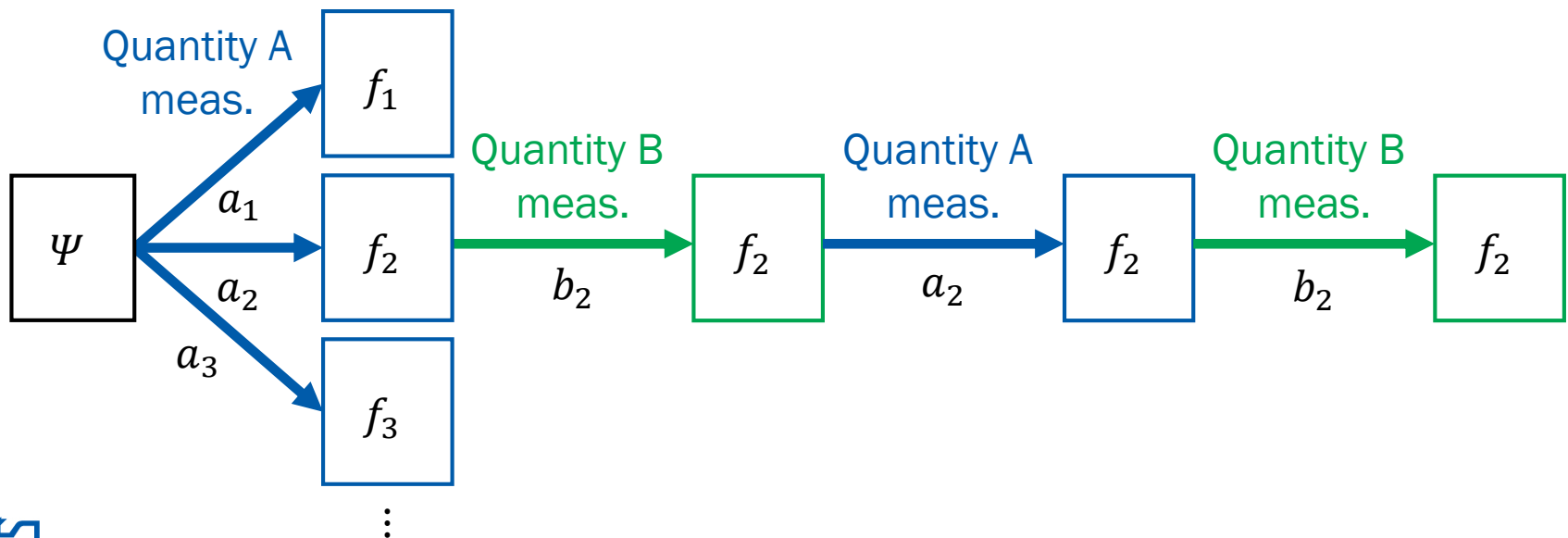
Schödinger's Cat



M 4.7 Measurement of Two Observables

If two QM operators share their eigenfunctions ...

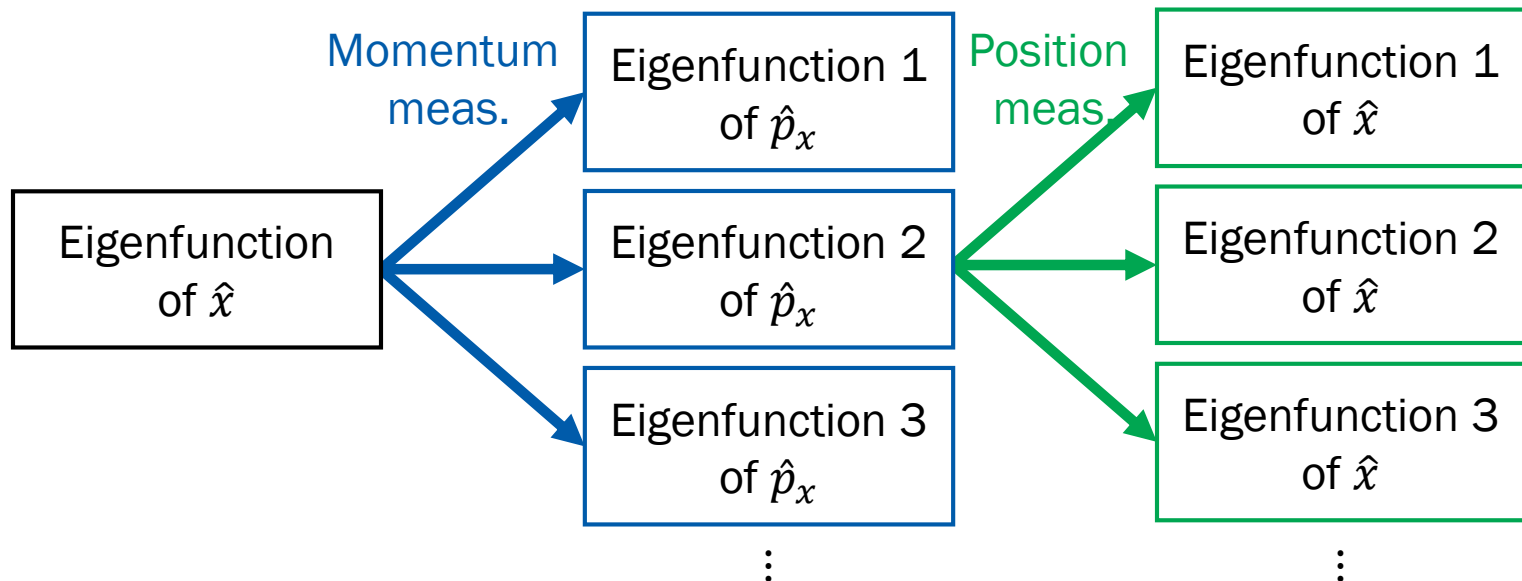
$$\hat{A}f_k = a_k f_k \text{ and } \hat{B}f_k = b_k f_k$$



M 4.4 Measurement of Two Observables

The eigenfunction of one QM operator may not be the eigenfunction of another QM operator

(e.g. eigenfunction of $\hat{x} \neq$ eigenfunction of $\hat{p}_x$)



Uncertainty Principle

For two QM operators that do not share the eigenfunctions,

It is impossible to simultaneously determine
the corresponding observables

- Position-momentum uncertainty (Gen Chem version)

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}$$

Review: Statistics

- Expected value

$$\langle g \rangle = \sum_{i=1}^{\infty} p_i g_i$$

- Variance

$$\sigma_g^2 = \langle (g - \langle g \rangle)^2 \rangle = \langle g^2 \rangle - \langle g \rangle^2$$

- Standard deviation

$$\sigma_g = (\sigma_g^2)^{1/2} = \sqrt{\langle g^2 \rangle - \langle g \rangle^2}$$

Expectation Value

For the eigenfunctions of the operator $\hat{A}$,

$$\hat{A}f_k = a_k f_k,$$

if the wavefunction of the system is

$$\Psi = c_1 f_1 + c_2 f_2 + c_3 f_3 + \cdots + c_n f_n,$$

the expectation value of the observable A is

$$\begin{aligned}\langle A \rangle &= a_1 P_1 + a_2 P_2 + \cdots + a_n P_n \\ &= a_1 |c_1|^2 + a_2 |c_2|^2 + \cdots + a_n |c_n|^2\end{aligned}$$

RS 2.6

M 4.3

Expectation Value

$$\langle A \rangle = a_1 |c_1|^2 + a_2 |c_2|^2 + \cdots + a_n |c_n|^2$$

Using the orthogonality (and normalization), we can show

$$\langle A \rangle = \int \Psi^* \hat{A} \Psi \, d\tau$$

Example 4-4

Calculate the average energy of the system in a state represented by

$$\Psi(x) = \left(\frac{30}{a^5}\right)^{1/2} x(a-x), \quad 0 \leq x \leq a$$

The average energy is the expectation value of energy:

$$\langle E \rangle = \int_0^a \Psi^*(x) \hat{H} \Psi(x) dx = \int_0^a \Psi^*(x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) \Psi(x) dx$$

$$\langle E \rangle = -\frac{15\hbar^2}{ma^5} \int_0^a x(a-x) \frac{d^2}{dx^2} [x(a-x)] dx$$

$$\langle E \rangle = \frac{30\hbar^2}{ma^5} \int_0^a x(a-x) dx = \frac{5\hbar^2}{ma^2}$$

Variance

$$\sigma_A^2 = \langle A^2 \rangle - \langle A \rangle^2$$

where

$$\langle A \rangle = \int \psi^* \hat{A} \psi \, d\tau$$

$$\langle A^2 \rangle = \int \psi^* \hat{A} \hat{A} \psi \, d\tau$$

- Standard deviation: $\sigma_A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$

Uncertainty Principle

For two QM operators that do not share the eigenfunctions,

It is impossible to simultaneously determine
the corresponding observables

- Position-momentum uncertainty (formal version)

$$\sigma_x \sigma_{p_x} \geq \frac{\hbar}{2}$$

Operator Multiplication

- Associative property

$$(\hat{A}\hat{B})\hat{C} = \hat{A}(\hat{B}\hat{C})$$

- Distributive property

$$\hat{A}(\hat{B} + \hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C}$$

- Commutative property

$$\hat{A}\hat{B} = \hat{B}\hat{A} ? \rightarrow \text{Not necessarily}$$

More on Commutative Property

- If $\hat{A}\hat{B} = \hat{B}\hat{A}$, $\hat{A}$ and $\hat{B}$ are said to **commute**
- **Commutator**

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If $\hat{A}$ and $\hat{B}$ commute,

$$[\hat{A}, \hat{B}] = \hat{0}$$

Generalized Uncertainty Principle

$$\sigma_A^2 \sigma_B^2 \geq -\frac{1}{4} \langle [\hat{A}, \hat{B}] \rangle^2$$

e.g. position-momentum uncertainty

$$[\hat{x}, \hat{p}_x] = \hat{x}\hat{p}_x - \hat{p}_x\hat{x} = x \frac{\hbar}{i} \frac{\partial}{\partial x} - \frac{\hbar}{i} \frac{\partial}{\partial x} x = i\hbar$$

$$\sigma_x^2 \sigma_{p_x}^2 \geq -\frac{1}{4} \langle i\hbar \rangle^2 = \left(\frac{\hbar}{2} \right)^2$$